JKUAT SODeL

JOMO KENYATTA UNIVERSITY OF AGRICULTURE & TECHNOLOGY

SCHOOL OF OPEN, DISTANCE AND eLEARNING
P.O. Box 62000, 00200
Nairobi, Kenya
E-mail: elearning@jkuat.ac.ke

ICS 2203 WEB APPLICATION DEVELOPMENT 1

LAST REVISION ON October 3, 2013

This presentation is intended to covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green Exercise tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elearning@jkuat.ac.ke.

LESSON 5

Web Hosting

Learning outcomes

Upon completing this topic, you should be able to:

1. Define Hosting.
2. Identify Hosting Types.
3. Relate Hosting and the considerations made in choice of a host.

5.1. Introduction

Let's start by defining a HOST.

Definition. In computer networking, a network host or Internet host or simply host is a computer connected to the Internet that can store shared information. A Host may also store client or server software.

Usually such a Computer is accessible by remotely by other computers. The term HOST is used when two computers are connected via modems, wireless networks or telephone lines. The Machine holding the data is the host and the one accessing is the terminal On the Internet every host has a unique IP address that includes a host address part. The host address may be assigned either manually by the computer administrator, or automatically by a Dynamic Host Configuration Protocol (DHCP) server.

Every host is also a network node (i.e. a network device), but every node is not a host. Network nodes such as modems and network switches are not assigned host addresses, hence are not considered as hosts. Devices such as network printers and hardware routers however are assigned IP host addresses, but since they are not general-purpose computers, they are sometimes not considered as hosts in the literature.

5.2. Why Hosting?

For any internet resource to be visible globally, it has to be hosted on a web server. Web servers are special applications that are responsible for delivering web pages requested for on the WWW. The communication protocol used to return the pages to the requesting machine is known as the HTTP(hyper text transfer protocol) Web hosting means storing your web site on a public server. Web hosting normally includes email services.

Web hosting often includes domain name registration.

5.2.1. How does the WWW work?

Web information is stored in documents called web pages. Web pages are hypertext files stored on computers called web servers. Computers reading the web pages are called web clients. Web clients view the pages with a program called a web browser. Popular browsers are Internet Explorer and Mozilla Firefox.

5.2.2. How does a Browser Fetch a Web Page?

A browser fetches a page from a web server by a request from a web client such as a web browser. Other examples of web client include mail client (Ms Outlook, or thunderbird), FTP clients(lime wire) e.t.c. A request is a standard HTTP request containing a page address at the URI bar i.e.

http://www.someone.com/page.htm.

5.2.3. How does a Browser Display a Web Page?

All web pages contain instructions for display The browser displays the page by reading these instructions. The most common display instructions are called HTML tags. HTML tags look like this <p>This is a Paragraph</p>.

5.2.4. What is an Internet Service Provider?

ISP stands for Internet Service Provider. An ISP provides Internet services. A common Internet service is web hosting. Web hosting means storing your web site on a public server. Web hosting normally includes email hosting among other services such as domain name registration. In choosing a host one has to make certain considerations to determine the best provider to suite ones needs.

Listed below are some of the hosting options one might consider.

5.2.5. Hosting Options

- **Hosting Your Own Web**

Considerations for doing your own hosting Include:

Hardware costs

- Servers are usually very expensive in terms of cost of specialized hardware, power backups and dedicated internet connection

Software costs

- Software licenses contribute to an added cost of hosting. Ranging from server software, anti-virus software, firewall e.t.c

Labour Costs

- These will be the costs associated with the maintenance of the site .ie. You have to deal with bugs and viruses, and keep your server constantly running in an environment.

5.2.6. Using Service Provider

Service providers have the following types of options for consideration; Hosting can be FREE, SHARED or DEDICATED.

Free Hosting

- Offered by a number of service providers freely on the web. These providers give limited space and bandwidth for the sites with the intention of placing advertisement on the sites that use this service.
- Similarly since the service is free the owners of the site will not be allowed to use the domain name of their choice. In most cases you will be given a false domain name.
- It is best suited for small sites with low traffic usually recreational sites or hobbies. They are however NOT recommended for business sites of site which attract high volumes of traffic.

- **Figure showing a free Hosting service provider**

Shared (Virtual) Hosting

- This is the most popular form of web hosting available. It is known for being very cost effective since it allows a number of websites to share a server and share in the cost implications.
- With shared hosting, your web site is hosted on a powerful server along with perhaps hundreds of other web sites. Unlike free hosting however each website will have its own domain name.
- The service provider may often offer multiple software solutions such as email, database, and back ups. Technical support is also provided.
- Despite these the shared hosting has a few challenges such

as compromised security due to many sires on the same server (i.e a virus attach will affect all on the same server), there is also limited support for the additional services including the amount of traffic provided by the service provide. This may vary from provider to provider.

Dedicated Hosting

- This involves hosting your site on a dedicated server. The server is completely dedicated to a specific website including support services for the server, internet connectivity, storage e.t.c.
- It is the most expensive form of hosting. The solution is best suited for large cooperate web sites with high traffic volumes, and web sites that use special software.

- This type of hosting has the benefit of security being guaranteed, powerful hardware and software is used hence it is more reliable.
- You should expect dedicated hosting to be very powerful and secure, with almost unlimited software solutions. The main deterrent to using this solution is the cost implications.

Collocated Hosting

- Collocation means "co-location". It is a solution that lets you place (locate) your own web server on the premises (locations) of a service provider.
- It is very similar to running your own server in your own office, only that it is located at a place better designed

for it that will come with the benefits of better internet speeds, security and monitoring services.

- Collocated Hosting is slightly expensive and requires technical know how on the part of the owner of the web sever.

5.2.7. Service Provider Options

In General the following are the considerations to look out for when shopping around for a service provider.

Connection Speed

- Find out the speeds your service provider has especially if you will be expecting high traffic volumes; most providers have very fast connections to the Internet, like full T3 fiber-optic 45Mps connections equivalent to about 2000 traditional (28K) modems or 1000 high speed (56K) modems.

Powerful Hardware

- Find out the minimum requirements for you website; Service providers often have many powerful web servers that can be shared by several companies. You expect them to have an effective load balancing, and backup servers if necessary.

Security and Stability

- Find out how they secure their websites; Most Internet Service Providers are specialists on web hosting. Expect their servers to have more than 99% up time, the latest software patches, and the best virus protection.

24-hour support

- Make sure your Internet service provider offers 24-hours support. Don't put yourself in a situation where you cannot fix critical problems without having to wait until the next working day. 24 hour Call services could be vital to reduce down time and unnecessary losses.

Daily Backup

- Make sure your service provider does a routine backup and follows proper back up policy to avoid loosing valuable data.

Traffic Volume

- Study the provider's traffic volume restrictions. Make sure that you don't have to pay a fortune for unexpected high traffic if your web site becomes popular.

Bandwidth or Content Restrictions

- Study the provider's bandwidth and content restrictions. If you plan to publish pictures or broadcast video or sound, make sure that you can.

Email Capabilities

- Make sure your provider fully supports the email capabilities you need. (You can read more about email capabilities in a later chapter).

Database Access

- Make sure your provider fully supports the database access you need if you plan to use databases from your site. (You can read more about database access in a later chapter).

5.2.8. Hosting and Domain Names

- A Domain Name is a unique name for your web site.
- Choosing a hosting solution should include domain name registration.
- Your domain name should be easy to remember and easy to type.

5.2.9. Registering a Domain

- Domains can be registered from domain name registration companies such as http://www.dotdnr.com.
- These companies provide interfaces to search for available domain names and they offer a variety of domain name extensions that can be registered at the same time.
- Domain Name Registration provides registration services for .com .net .org .biz .info .us .nu .ws .cc and .tv domains.
- Newer domain extensions such as .biz .info and .us have more choices available as many of the popular domains have yet to be taken. While .com and .net domains are well established and recognized, most popular domains with these extensions are already registered.

◀◀ ▶▶
◀ ▶
◀ Doc Doc ▶
Back Close

5.2.10. Choosing Your Domain

- Choosing a domain is a major step for any individual or organization.
- While domains are being registered at a record, new domain extensions and creative thinking still offer thousands of excellent choices. When selecting a name it is important to consider the purpose of a domain name, which is to provide people an easy way to reach your web site. The best domains have the following characteristics:
- Short - People don't like to type! The shorter your domain, the easier it is to reach and the less are the chance the user will make a typographical error while typing it.
- Meaningful - A short domain is nothing without meaning,

34i4nh69.com is only 8 characters long but would not be easy to enter or remember. Select a domain that relates to your site in a way that people will understand.

- Clear - Clarity is important when selecting a domain name. You should avoid selecting a name that is difficult to spell or pronounce. Also, pay close attention to how your domain sounds and how effectively it can be communicated over the phone.
- Exposure: Just like premium real-estate on the ground that gets the most exposure, names that are short and easy to remember are an asset. In addition to humans viewing your domain, you should consider search engines. Search engines index your site and rank it for relevance against terms people search for online. In order to maximize your

sites exposure, consider including a relevant search term in your domain. Of course, this should only be considered if it still maintains a short, clear and meaningful domain.

5.2.11. Sub Domains

- Most people are unaware but they already use sub domains on a daily basis. The famous "www" of the World Wide Web is the most common example of a sub domain. Sub domains can be created on a DNS server and they don't need to be registered with a domain registrar, of course, the original domain would need to be registered before a sub domain could be created. Common examples of sub domains used on the internet are http://store.apple.com and http://support.microsoft.com.

- Sub domains can be requested from your wConnection Speed
- Find out the speeds your service provider has especially if you will be expecting high traffic volumes; most providers have very fast connections to the Internet, like full T3 fiber-optic 45Mps connections equivalent to about 2000 traditional (28K) modems or 1000 high speed (56K) modems.

Powerful Hardware

- Find out the minimum requirements for you website; Service providers often have many powerful web servers that can be shared by several companies. You expect them to have an effective load balancing, and backup servers if necessary.

Security and Stability

- Find out how they secure their websites; Most Internet Service Providers are specialists on web hosting. Expect their servers to have more than 99% up time, the latest software patches, and the best virus protection.

24-hour support

- Make sure your Internet service provider offers 24-hours support. Don't put yourself in a situation where you cannot fix critical problems without having to wait until the next working day. 24 hour Call services could be vital to reduce down time and unnecessary losses.

Daily Backup

- Make sure your service provider does a routine backup and follows proper back up policy to avoid loosing valuable data.

Traffic Volume

- Study the provider's traffic volume restrictions. Make sure that you don't have to pay a fortune for unexpected high traffic if your web site becomes popular.

Bandwidth or Content Restrictions

- Study the provider's bandwidth and content restrictions. If you plan to publish pictures or broadcast video or sound, make sure that you can.

◀◀ ▶▶ ◀ ▶ ◀ Doc Doc ▶ Back Close

Email Capabilities

- Make sure your provider fully supports the email capabilities you need. (You can read more about email capabilities in a later chapter).

Database Access

- eb hosting provider or created by yourself if you manage your own DNS server.

5.2.12. False Domain Names - Directory Listings

- Some providers will offer you a unique name under their own name like: www.theircompany.com/yourcompany/
- This is not a real domain name, it is a directory - and you should try to avoid it.

- These URLs are not desirable, especially for companies. Try to avoid them if you can afford to register a domain. Typically these are more commonly used for personal sites and free sites provided by your ISP, you may have seen www.theircompany.com/~username as a common address, this is just another way to share a single domain and provide users their own address.
- Open competition in domain name registration has brought about a dramatic decrease in pricing so domain sharing is far less common since people can register their own domains for only $15 per year.

5.2.13. Expired Domains

- Another source for domain registrations is expired domains. When you register a domain, think of it as a rental, assuming there are no legal or trademark issues with the domain name, you are free to use it as long as you continue to pay the yearly fee (you can now also register in advance as many as 10 years). Some people register domains as speculators, hoping that they can later sell them, while others may have planned to use a domain and never had the time. The result is that domains that were previously registered regularly become available for registration again. You can see, and search through a list of recently expired domains for free at http://www.dotdnr.com. If you wish to register an expired domain you pay the same fee as you would for a new registration.

5.2.14. Use Your Domain Name

- After you have chosen - and registered - your own domain name, make sure you use it on all your web pages and on all your correspondence, like email and traditional mail.
- It is important to let other people be aware of your name, and to inform your partners and customers about your web site.

5.2.15. Hosting Capacities

Make sure you get the disk space and the traffic volume you need.

- **How Much Disk Space?**
  - small or medium web site will need between 10 and 100MB of disk space.
  - If you look at the size of HTML pages, you will see that the average size is very small. Maybe even smaller than 1KB. But if you look at the size of the images (button, gif, banner, jpg) used inside the pages, you will often find images many times larger than the page itself.
  - Expect each HTML page to take up between 5 and 50KB of disk space on your web server, depending on the use of images or other space consuming elements.
  - If you plan to use lots of images or graphic elements (not to mention sound files or movies), you might be needing

much more disk space.

- Make sure that you know your needs before you start looking for your web host.

- **Monthly Traffic.**
  - A small or medium web site will consume between 1GB and 5GB of data transfer per month.
  - You can calculate this by multiplying your average page size with the number of expected page views per month. If your average page size is 30KB and you expect 50,000 page views per month, you will need 0.03MB x 50,000 = 1.5GB.
  - Larger, commercial sites often consume more than 100GB of monthly traffic.

- Before you sign a contract with a host provider, make sure to check this:
- What are the restrictions on monthly transfer
- Will your site be closed if you exceed the volume
- Will you be billed a fortune if you exceed the volume
- Will my future need be covered
- Is upgrading a simple task

5.2.16. Connection Speed

- Visitors to your web site will often connect via a modem, but your host provider should have a much faster connection.

- In the early days of the Internet a T1 connection was considered a fast connection. Today connection speeds are much faster.
- 1 byte equals to 8 bits (and that's the number of bits used to transport one character). Low speed communication modems can transport from about 14 000 to 56 000 bits per second (14 to 56 kilobits per second). That is somewhere between 2000 and 7000 characters per second, or about 1 to 5 pages of written text.
- One kilobit (Kb) is 1024 bits. One megabit (Mb) is 1024 kilobits. One gigabit (Gb) is 1024 megabits.
- Before you sign up a contract with any hosting provider, surf some other web sites on their servers, and try to get a good feeling about their network speed. Also compare the

other sites against yours, to see if it looks like you have the same needs. Contacting some of the other customers is also a valuable option.

5.2.17. Web Hosting Server Technologies

This section describes some of the most common hosting technologies.

- **Windows Hosting**
  - Windows hosting means hosting of web services that runs on the Windows operating system.
  - You should choose Windows hosting if you plan to use ASP (Active Server Pages) as server scripting, or if you plan to use a database like Microsoft Access or Microsoft

SQL Server. Windows hosting is also the best choice if you plan to develop your web site using Microsoft Front Page.

- **Unix Hosting**
  - Unix hosting means hosting of web services that runs on the Unix operating system.
  - Unix was the first (original) web server operating system, and it is known for being reliable and stable. Often less expensive than Windows.
- **Linux Hosting**
  - Linux hosting means hosting of web services that runs on the Linux operating system.

5.2.18. Server Side Technologies

- **CGI**
  - Web pages can be executed as CGI scripts. CGI scripts are executables that will execute on the server to produce dynamic and interactive web pages.
  - Most Internet service providers will offer some kind of CGI capabilities. Internet service providers will often offer pre-installed, ready to run, guest-books, page-counters, and chat-forums solutions written in CGI scripts.
  - The use of CGI is most common on Unix or Linux servers.
- **ASP - Active Server Pages**
  - Active Server Pages is a server-side scripting technology developed by Microsoft.

- With ASP you can create dynamic web pages by putting script code inside your HTML pages. The code is executed by the web server before the page is returned to the browser. Both Visual Basic and JavaScript can be used.
- ASP is a standard component in Windows 95,98, 2000, and XP. It can be activated on all computers running Windows.

- **PHP – Personal Home Pages**
  - Active Server Pages is a server-side scripting technology developed by Microsoft.
  - With ASP you can create dynamic web pages by putting script code inside your HTML pages. The code is executed by the web server before the page is returned to the browser. Both Visual Basic and JavaScript can be used.

- ASP is a standard component in Windows 95,98, 2000, and XP. It can be activated on all computers running Windows.

- **PHP – Personal Home Pages**
  - PHP is an open-source software and is a powerful server-side scripting language for creating dynamic and interactive websites.
  - PHP is the widely-used, free, and efficient alternative to competitors such as Microsoft's ASP. PHP is perfectly suited for Web development and can be embedded directly into the HTML code.
  - The PHP syntax is very similar to Perl and C. PHP is often used together with Apache (web server) on various operating systems. It also supports ISAPI and can be used

with Microsoft's IIS on Windows.

- PHP supports many databases, such as MySQL, Informix, Oracle, Sybase, Solid, PostgreSQL, Generic ODBC, etc.

- **JSP- Java Server Pages**
  - JSP is a server-side technology much like ASP developed by Sun.
  - With JSP you can create dynamic web pages by putting Java code inside your HTML pages. The code is executed by the web server before the page is returned to the browser.
  - Since JSP uses Java, the technology is not restricted to any server-specific platform.

- **Cold Fusion**
  - Cold Fusion is another server-side scripting language used to develop dynamic web pages.
  - Cold Fusion is developed by Macromedia.

5.2.19. Web Hosting Database Technologies

- MS SQL Server or Oracle for high traffic database-driven web sites.
- MySQL for low-cost database-access.
- MS Access for low traffic web sites.

5.2.20. Web Databases

- If your web site needs to update large quantities of information via the web, you will need a database to store your information.

- There are many different database systems available for web hosting. The most common are MySQL, SQL Server, Oracle, and MS Access.

5.2.21. Using the SQL Language

- SQL is the language for accessing databases.
- If you want your web site to be able to store and retrieve data from a database, your web server should have access to a database-system that uses the SQL language.

5.2.22. SQL Server

- Microsoft's SQL Server is a popular database software for database-driven web sites with high traffic.

- SQL Server is a very powerful, robust and full featured SQL database system.

5.2.23. Oracle

- Oracle is also a popular database software for database-driven web sites with high traffic.
- Oracle is a very powerful, robust and full featured SQL database system.

5.2.24. MySQL

- MySQL is also a popular database software for web sites.
- MySQL is a very powerful, robust and full featured SQL database system.

- MySQL is an inexpensive alternative to the expensive Microsoft and Oracle solutions.

5.2.25. Access

- When a web site requires only a simple database, Microsoft Access can be a solution.
- Access is not well suited for very high-traffic, and not as powerful as MySQL, SQL Server, or Oracle.

In Summary Your Web Hosting Check List

Before you choose your web host, make sure that:

- The hosting type suits your current needs, i.e. The hosting type is cost effective.
- In case of a service provide, You Know the details about the services they provide (speeds, software e.t.c)

- You are allowed to Upgrade or Downgrade when necessitated.
- Before you sign up a contract with any hosting provider, surf some other web sites on their servers, and try to get a good feeling about their network speed. Also compare the other sites against yours, to see if it looks like you have the same needs. Contacting some of the other customers is also a valuable option.

Example ✎. What should be in your web hosting checklist

Solution: Before you choose your web host, make sure that:

- The hosting type suits your current needs, i.e. The hosting type is cost effective.

- In case of a service provide, You Know the details about the services they provide (speeds, software e.t.c)
- You are allowed to Upgrade or Downgrade when necessitated.
- Before you sign up a contract with any hosting provider, surf some other web sites on their servers, and try to get a good feeling about their network speed. Also compare the other sites against yours, to see if it looks like you have the same needs. Contacting some of the other customers is also a valuable option.

□

Learning Activities

Revision Questions

EXERCISE 1. ✍ Define a web host .

EXERCISE 2. ✍ List the Different types of Hosting .

EXERCISE 3. ✍ Highlight the considerations you would have to make when choosing a web host

References and Additional Reading Materials

- George Beekman,(2008) Tomorrow's Technology and You, Complete, 9/E,Oregon State University Ben Beekman, ISBN-10: 0135045045 • ISBN-13: 9780135045046 ©2010 • Prentice Hall • Paper, 672 pp

- Chuck Musciano and Bill Kennedy(2000)HTML & XHTML: The Definitive Guide, 4th Edition, Oreilly, ISBN 0-596-00026-X
- Craig Grannell(2007 The Essential Guide to CSS and HTML Web Design, Springer, ISBN-13 (pbk): 978-1-59059-907-5

Solutions to Exercises

Exercise 1. This is a Machine/computer on a network or internet that shores a resource that is shared or is required by other machines Exercise 1

Exercise 2. Shared Hosting, Dedicated Hosting, Co located Hosting, Free Hosting Exercise 2

Exercise 3.

Connection Speed, Powerful Hardware, Security and Stability, 24-hour support, Daily Backup, Traffic Volume, Bandwidth or Content Restrictions, Email Capabilities, Database Access

Exercise 3

◀◀ ▶▶ ◀ ▶ ◀ Doc Doc ▶ Back Close